NICFW TD-H3 25 Channel Editor

Android App — Complete User Guide

github.com/jnyer27/NICFW-H3-25-CHIRP-ADAPTER

CHIRP-compatible · USB OTG · Android

TD-H3 Channel Editor — User Guide

> ***App:*** *NICFW TD-H3 Channel Editor (**Android v2.2.0** — see release notes and app CHANGELOG)*
> ***Radio:*** *TIDRadio TD-H3 running **nicFW v2.5** **Platform:** Android (min SDK 24 / Android 7.0)*

Documentation: Browse this guide on **GitHub Pages** or download the live PDF at **user-guide.pdf**. Every **GitHub Release** gets the same PDF as `nicfw-td-h3-editor-<release-tag>-userguide.pdf` plus a **User guide** section in the release notes (MkDocs URL + PDF links), maintained by the **Update User Guide** workflow when the guide is rebuilt or releases are backfilled.

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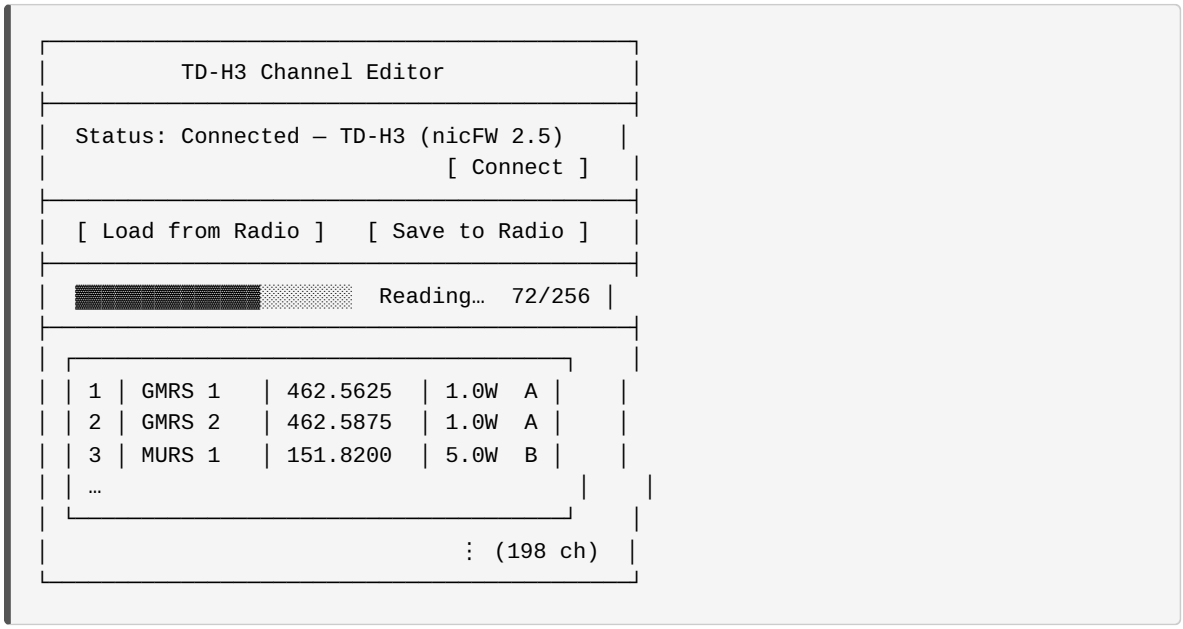
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1. Overview

The **TD-H3 Channel Editor** is an Android companion app for the TIDRadio TD-H3 running nicFW v2.5 firmware. It communicates with the radio over **Bluetooth** (BLE or Classic SPP) to read and write the full 8 KB EEPROM — the same memory block that stores all channels, band plan entries, scan presets, and group labels.

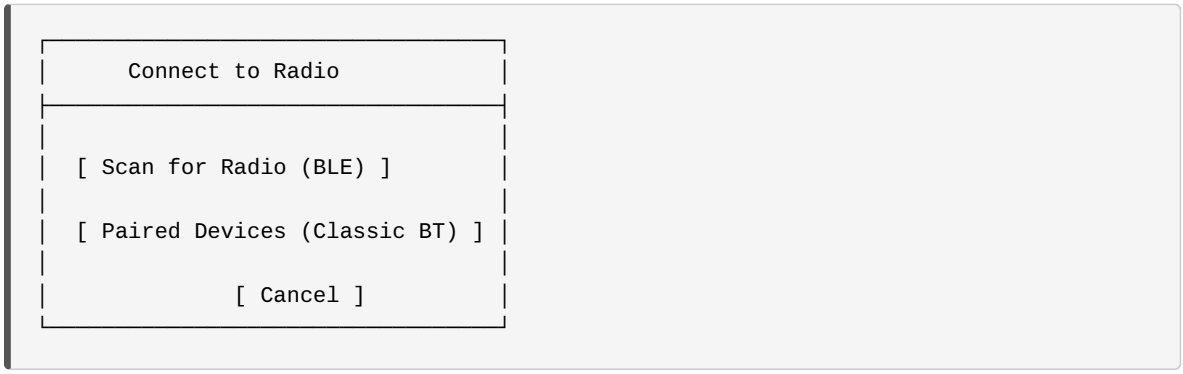


Key concept: All edits are made in memory only. Nothing changes on the radio until you tap **Save to Radio**. This lets you make multiple changes and upload them all at once.

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2. Connecting to the Radio

Tap **[Connect]** on the main screen. A dialog appears with two options:



Option A — BLE (Recommended)

As of app v1.1.0, BLE discovery is tuned for reliability and for common BLE-UART bridges, not only the TD-H3's built-in radio.

- Tap **Scan for Radio (BLE)**.
- The list shows devices whose **advertising packet includes a known UART service UUID** (nicFW `0000ff00-…`, Nordic UART, HM-10/TI `0000ffe0-…`, Microchip/ISSC, etc.). Your TD-H3 usually appears as **TD-H3** or similar; third-party serial dongles that

advertise one of those services also appear.

- Tap the device name to connect. The app negotiates a **moderate BLE MTU** (with a safe **20-byte fallback**) so cheap adapters are less likely to drop the link than with an aggressive 512-byte request; EEPROM traffic over the air is unchanged from older app versions.
- Android 12+ requires the **Nearby Devices** permission; Android 11 and below requires **Location** permission. Grant when prompted.

For a concise technical list of changes, see **AndroidNICFW_CH_EDITOR/CHANGELOG.md** (section **1.1.0**).

Option B — Classic Bluetooth (SPP) *(unconfirmed)*

> ***Note:*** *Classic SPP connectivity has not been confirmed to work with the TD-H3 and nicFW 2.5. BLE is the recommended and verified connection method.*

- Pair the radio to your phone first via Android's system Bluetooth settings.
- Tap **Paired Devices (Classic BT)** and select the radio from the list.

Connection Status

Status indicator	Meaning
`Status: Disconnected`	No radio connected
`Status: Connecting…`	Handshake in progress
`Status: Connected — <name>`	Ready to load/save

> ***Tip:*** *The BLE connection uses nicFW's remote-control protocol (0x4A enable / 0x4B disable). If the radio firmware does not support remote mode, use Classic BT. If* ***BLE scan is empty*** *but the radio is on, the handset may not be advertising a recognized UART UUID — use* ***Paired Devices (Classic BT)*** *or confirm firmware/advertising (v1.1.0 scan filters; see CHANGELOG above).*

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3. Loading & Saving the Radio Memory

Load from Radio

Tap **[Load from Radio]** (only enabled when connected). The app reads the full 8 KB EEPROM in 256 × 32-byte blocks. A progress bar and block counter are shown during the transfer (about 10–30 seconds depending on connection speed).

```
[ Load from Radio ]   [ Save to Radio ]
▓▓▓
```

- **Groups (`GRP:*`)** = dark gray.
- **Power (`PWR:*`)** = gray (N/T or 0), yellow (1–70), red (>70).
- **Mode** = color-coded by value (`FM`, `AM`, `USB`, `AUTO`) with the same chip style.
- **Duplex** = color-coded by value (`Simplex`, `+xMHz`, `-xMHz`, `Split`) with the same chip style.
- **Bandwidth** = blue (BW:W), orange (BW:N).
- **Tone chips** = cyan for CTCSS, magenta for DCS.

Column descriptions:

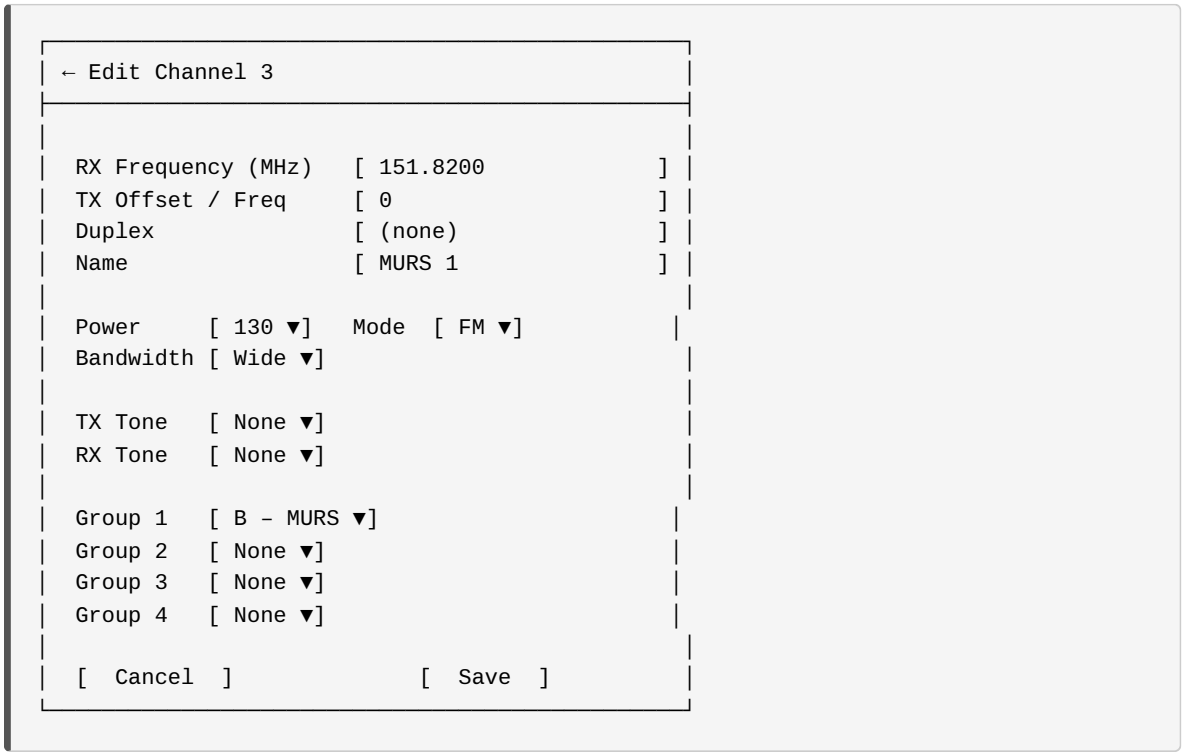
Column	Description
`#`	Channel slot number (1–198, fixed), shown top-left of each row
Frequency	RX frequency shown directly under channel number
Groups	Group chips shown under frequency in the left column
Name	Channel name shown on the right, top line, larger text
Chips	Power (`PWR:`), Mode (`FM/AM/USB/AUTO`), Duplex (`Simplex`, `+xMHz`, `-xMHz`, `Split`), Bandwidth (`BW:W` / `BW:N`), and conditional Tone chips (`TX:` / `RX:`)
Chip colors	Power: gray (`N/T/0`), yellow (`1–70`), red (`>70`); bandwidth: blue/orange; duplex and mode are color-coded by value

- **Tap** a channel row to open the channel editor.
- **Long-press** a channel row to enter Multi-Select Mode.
- **Drag** a channel by its drag handle (≡) to reorder within the list. Drag reordering writes new slot numbers into the EEPROM immediately in memory.

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5. Editing a Single Channel

Tap any channel row to open the **Channel Editor**.



Field Reference

Field	Valid Values	Notes
RX Frequency	MHz, 4 decimal places	Leave blank to mark slot as empty
TX Offset / Freq	kHz (duplex ±) or MHz (split)	Depends on Duplex setting
Duplex	(blank), `+`, `-`, `split`	`+` adds offset, `-` subtracts, `split` sets independent TX
Name	Up to 12 characters	Stored as ASCII in EEPROM
Power	N/T, 1–255 (raw byte)	See power-to-watt table below
Mode	Auto, FM, AM, USB	
Bandwidth	Wide, Narrow	
Busy Lock	On / Off	When On, prevents accidental TX on an occupied channel. Automatically forced Off and disabled whenever a TX offset or duplex mode is set — incompatible with repeater/split channels.

	754 N/R	Flat picker — all tones in one list
RX Tone	Same as TX Tone	Squelch decode tone
Group 1–4	None, A–O (with custom labels)	Up to 4 groups per channel

Power Calibration

The power spinner stores a raw byte (0–255). Approximate watt equivalents:

Raw value	Approx. watts
0 / N/T	No transmit
29	0.5 W
58	2.0 W
130	5.0 W
255	~8.7 W (extrapolated)

> *Values between calibration points are linearly interpolated. The watt label shown in the channel list is an estimate; actual output depends on battery and frequency.*

Power Cap Advisory

If **Tune Settings** has a VHF or UHF power cap configured (see §8.8), a non-blocking warning appears immediately below the Power spinner when the selected value exceeds the applicable cap:

```
⚠ Exceeds VHF cap (130 ≈ 5.0W) — radio will clamp to cap at TX time
```

The spinner is **not restricted** — you may store any value. The radio enforces the cap silently at transmit time; the EEPROM byte is unchanged. This lets you retain a high stored power value and simply raise the cap later without re-editing every channel.

Tone Picker Format

The flat tone spinner uses these formats:

Display	Type
`None`	No tone
`CTCSS 100.0 Hz`	CTCSS sub-audible tone
`DCS 023 N`	DCS code 023, Normal polarity

`DCS 023 R`	DCS code 023, Reverse polarity

Saving / Clearing a Channel

- Tap **Save** to write changes into the in-memory EEPROM buffer.
- To **clear** a channel slot (make it empty), delete the RX Frequency field and tap Save.
- Changes take effect on the radio only after **Save to Radio** from the main screen.

---

6. Multi-Select Mode

Long-press any channel row to activate multi-select mode. A colored action bar appears at the bottom of the screen:

```
┌──────────────────────────────────────────────────────────┐
│  Channel List                                            │
│  …                                                       │
│ ✓ | GMRS 1  | 462.5625  |  1.0W |  A  | N  ← selected |
│ ✓ | GMRS 2  | 462.5875  |  1.0W |  A  | N  ← selected |
│   | MURS 1  | 151.8200  |  5.0W |  B  | N             |
│ ✓ | GMRS 3  | 462.6125  |  1.0W |  A  | N  ← selected |
│  …                                                       │
├──────────────────────────────────────────────────────────┤
│ 3 selected  | ↑ | ↓ | ⇊ | ✎ | ↑□ | 🗑 | Done |
└──────────────────────────────────────────────────────────┘
```

Selection bar buttons:

Button	Icon	Action
Move Up	↑	Shifts all selected channels one slot up (swaps with the channel above)
Move Down	↓	Shifts all selected channels one slot down
Move to Slot	⇊	Opens a slot picker — moves the selected block to any slot (1–198) in one step
Bulk Edit	✎	Opens a field picker (TX power, groups, modulation, bandwidth, busy lock)
Export CSV	↑□	Names and exports selected channels as a CHIRP-compatible CSV file
Delete	🗑	Clears (empties) all selected channel slots
Done	text	Exits multi-select mode, returns to normal browsing

- Tap additional rows while in multi-select mode to add/remove them from the selection.
- Tap a selected row again to deselect it.
- The selection count in the bar updates in real time.

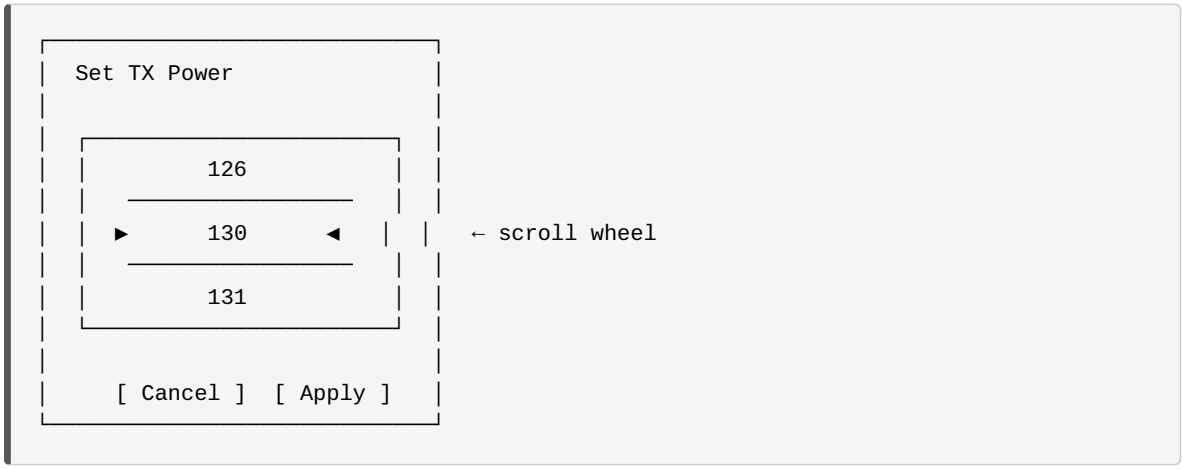
6.1 Move Up / Down

Move Up (↑) and Move Down (↓) shift the selected channels one position at a time in the channel list. The swap is performed in memory; tap **Save to Radio** afterward to persist.

> ***Tip:*** *For large moves, use* ***Move to Slot*** *(⤓) to jump the selection directly to any slot in one step. For group-based reordering, use* ***Sort Channels by Group*** *from the overflow menu.*

6.2 Set TX Power (Bulk)

Tap **Bulk Edit** (✎), then choose **TX power**:



- The scroll wheel (NumberPicker) is pre-seeded to the current power of the **first selected non-empty channel**.
- Scroll the wheel to pick any value: `N/T`, `1`–`255`.
- Tap **Apply** to write the chosen power to every selected non-empty channel.
- Empty slots in the selection are skipped.

Power values at a glance:



Power Cap Advisory

If the selected channels include VHF or UHF frequencies and a power cap is set in **Tune Settings** (see §8.8), an advisory appears **below the scroll wheel** when the chosen value exceeds the applicable cap:

```
⚠ Exceeds VHF cap (130 ≈ 5.0W) — radio will clamp to cap at TX time
```

When the selection spans both bands, the **more restrictive** of the two caps is used in the advisory. Apply is never blocked — the stored value is saved as-is.

6.3 Set Channel Groups (Bulk)

Tap **Bulk Edit** (✎), then choose **Channel groups**:

```
┌──────────────────────────────────────────┐
│  Set Groups for Selected Channels        │
├──────────────────────────────────────────┤
│                                          │
│  Group Slot 1  [ — Keep — ▼]             │
│  Group Slot 2  [ A — GMRS ▼]             │
│  Group Slot 3  [ — Keep — ▼]             │
│  Group Slot 4  [ — Keep — ▼]             │
│                                          │
│          [ Cancel ]  [ Apply ]           │
└──────────────────────────────────────────┘
```

Spinner options for each group slot:

Option	Meaning
`— Keep —`	Do not change this slot on the selected channels
`None`	Clear this slot (remove group assignment)
`A` through `0` (with labels)	Assign the displayed group

- Each slot spinner is **independent** — you can set slot 2 to `A — GMRS` while leaving slots 1, 3, and 4 unchanged (keep their current values).
- Custom group names (e.g., "GMRS", "MURS") are shown inline from the **Group Labels** you defined in Edit Group Labels.
- Tap **Apply** to bulk-assign groups to all selected non-empty channels.

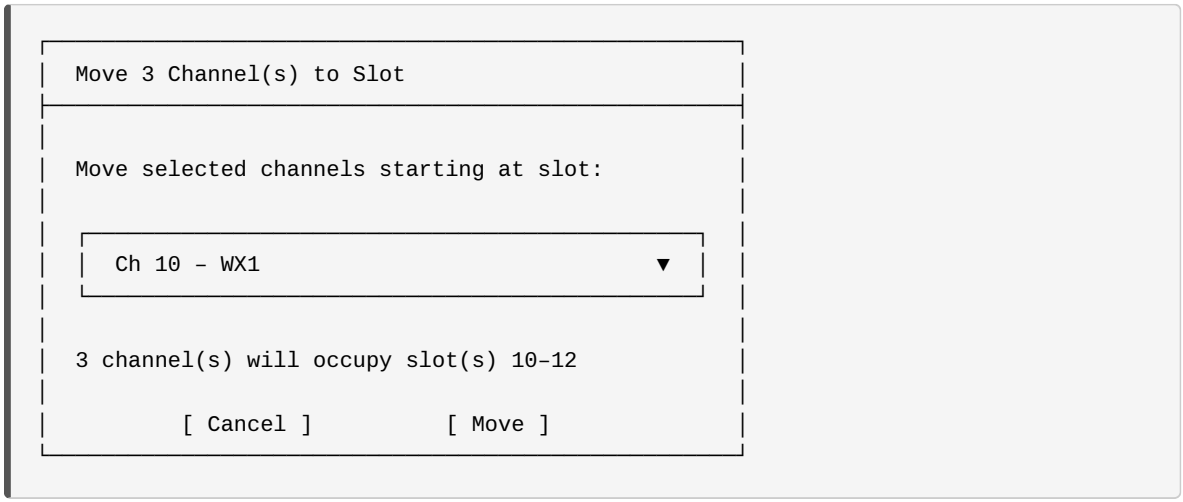
6.4 Delete / Clear Selected

Tap the 🗑 (delete) button to clear all selected channels. A confirmation dialog appears showing the count. Clearing sets those slots to empty (all bytes zeroed).

> *Empty slots appear as `(empty)` in the channel list and can be reused by importing CHIRP CSV data or by editing them manually.*

6.5 Move to Slot

Tap the ⬇ (arrow-to-underline) button while channels are selected to jump the entire selection to any slot in one action — no need to tap Move Up / Down dozens of times.



How it works:

1. Long-press to enter multi-select mode, then tap all channels you want to move.
2. Tap ⬇ — the dialog opens with the Spinner pre-set to the first selected slot.
3. Scroll the Spinner to choose the target **starting slot** (1–198).
4. The hint line updates live: *"N channel(s) will occupy slot(s) X–Y"*.
5. Tap **Move** — the selected channels are removed from their current positions and inserted starting at the chosen slot. All remaining channels shift to fill the gaps.

Selection after move:
The moved channels remain selected (highlighted) in their new positions so you can immediately continue editing or move them again.
Edge cases:

Situation	Result

Target slot is inside the current selection range	Computed from non-selected channels only — result is predictable
All channels selected	Early exit with a toast — nothing to move relative to
Target slot > available slots after removals	Channels land at the end of the list

---

6.6 Export as CHIRP CSV

Tap the ↑□ (upload arrow) button while channels are selected to export the selection to a CHIRP-compatible CSV file and share it.

```
┌──────────────────────────────────────────────┐
│  Export CHIRP CSV                            │
│  12 channel(s) will be exported.             │
│  File name:                                  │
│  ┌──────────────────────────────────────┐    │
│  │  chirp_export_20260313_104500        │    │
│  └──────────────────────────────────────┘    │
│          [ Cancel ]      [ Export & Share ]  │
└──────────────────────────────────────────────┘
```

1. Long-press to enter multi-select mode and select the channels you want to export.
2. Use **Channel Search** (§7) to quickly select a whole group.
3. Tap the ↑□ icon in the selection bar.
4. Edit the file name if desired (default is `chirp_export_<timestamp>`).
5. Tap **Export & Share** — the Android share sheet opens so you can send the file via email, cloud storage, AirDrop, etc.

The exported CSV: - Is numbered sequentially from `Location 0` regardless of original slot numbers. - Preserves frequency, name, duplex/offset, tone (CTCSS/DCS), mode, bandwidth, and power. - Can be imported directly back into the app (⋮ → **Import CHIRP CSV**) or into the desktop CHIRP software. - Empty slots in the selection are automatically skipped.

> ***Typical use case:*** *Select all channels in group H–GMRS using Search → Select All, then export to share your repeater list with another user.*

---

7. Channel Search

Tap the 🔍 (magnifying glass) icon in the top toolbar to open the search bar.

```
┌──────────────────────────────────────────────────┐
│  TD-H3 Channel Editor                  🔍  ⋮   │
├──────────────────────────────────────────────────┤
│  [ Search by name, group or frequency… ] [Select All] [×] │
├──────────────────────────────────────────────────┤
│  → Ch 31  Towson WRJY6     462.6000 MHz          │
│  → Ch 32  Baltimore WR     462.6750 MHz          │
│  → Ch 35  Lanham WQMM8     462.5500 MHz          │
│  …                                               │
└──────────────────────────────────────────────────┘
```

How it works: - Type any part of a **channel name**, **group label**, or **RX frequency** — the list filters live as you type (case-insensitive). - Matching is on name OR frequency OR any of the channel's group labels: - Typing `GMRS` shows channels assigned to groups whose label contains "GMRS". - Typing `boat` shows channels with "Boat" in the name or assigned to a group labeled "Boat". - Typing `462.5` shows all channels whose RX frequency starts with 462.5 MHz (e.g. 462.5000, 462.5125, 462.5625, ...). - Typing `462` shows every channel in the 462 MHz band. - Empty slots are always hidden while a search query is active. - Tap ✕ to clear the search and restore the full channel list. - Tap the 🔍 icon again to close the search bar. **Select All Matches:** Once you have results, tap the □ (hollow-square select-all icon) to enter multi-select mode with every visible (filtered) channel pre-selected. From there you can: - Tap **Export CSV** (↑□) to export the whole matched set as a CHIRP CSV. - Tap **Bulk Edit** (✎) then **TX power** to change power for the whole group at once. - Tap **Bulk Edit** (✎) then **Channel groups** to bulk-assign groups to all matches. - Tap any card to deselect individually, or tap **Done** to exit selection mode.

> ***Example workflow — share all GMRS repeaters:*** *1. Tap 🔍, type* `GMRS`*. 2. Tap □ — all channels in your GMRS group are selected. 3. Tap ↑□ → name the file →* ***Export & Share****.*

---

8. Overflow Menu Features

Tap the ⋮ (three-dot) menu in the top-right toolbar to access advanced features.

```
┌──────────────────────────────────────┐
│ ⋮                                    │
│  Import CHIRP CSV from File…         │
│  Import CHIRP CSV from Clipboard…    │
│  Sort Channels by Group…             │
│  Edit Group Labels…                  │
│  Save EEPROM dump…                   │
│  Import EEPROM dump…                 │
│  Edit Band Plan…                     │
│  Edit Scan Presets…                  │
│  Radio Settings…                     │
│  Tune Settings…                      │
│  XTAL 671 Calculator…                │
│  ✓ Protect Tune Settings             │
└──────────────────────────────────────┘
```

> ***XTAL 671 Calculator*** *is always accessible — no EEPROM required.* ***Import EEPROM dump*** *is always accessible regardless of connection state.* ***Protect Tune Settings*** *is a checkable toggle — accessible any time; state persists across app restarts. All other items require an EEPROM to be loaded first.*

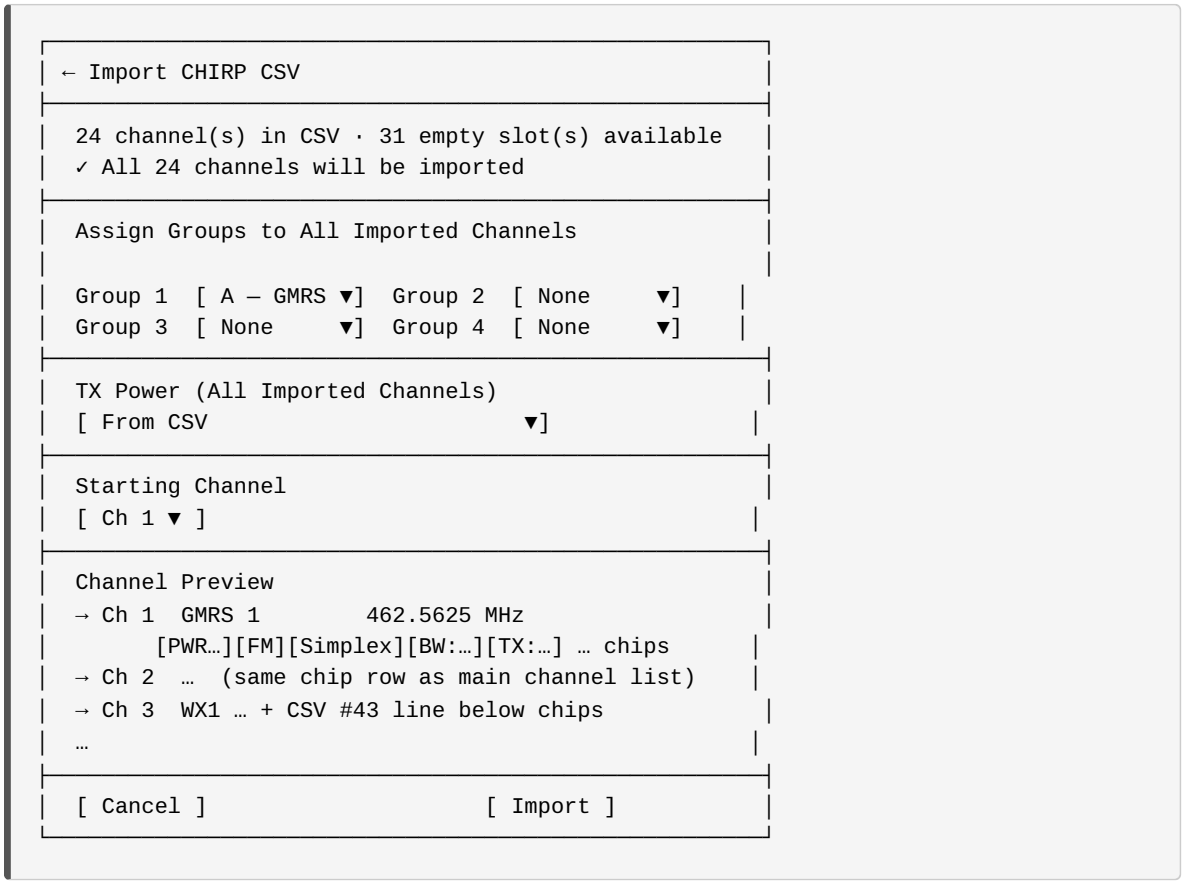
8.1 Import CHIRP CSV

Import a frequency list exported by the CHIRP radio programming software.
Two input methods are supported:

- ⋮ → **Import CHIRP CSV from File…** — opens the system file picker to select a `.csv` file.
- ⋮ → **Import CHIRP CSV from Clipboard…** — reads a CHIRP CSV copied directly to the clipboard.

Steps

1. Export your channel list from CHIRP as a CSV file (`File → Export`).
2. Transfer the CSV to your Android device (email, cloud storage, USB file transfer, etc.), **or** copy the CSV text to the clipboard on a device where CHIRP is running.
3. In the app, tap ⋮ → **Import CHIRP CSV from File…** and pick the file, **or** tap ⋮ → **Import CHIRP CSV from Clipboard…** to import from the clipboard.
4. The **CHIRP Import** screen opens:



1. Optionally assign all imported channels to up to 4 groups using the **Group** spinners.
2. Use the **TX Power** spinner to set a uniform power level for every imported channel:
3. **From CSV** *(default)* — keeps whatever power value the CSV row contained.
4. **N/T** — no transmit (useful for receive-only or monitor channels).
5. **1–255** — raw power byte; see the Power Calibration table in §5 for approximate watt equivalents.
6. Use the **Starting Channel** spinner to choose which empty slot the first imported channel lands in. The preview updates live — the `→ Ch N` labels shift to reflect your selection. Only positions where all channels fit without overwriting occupied slots are listed.
7. Tap **Import** to write channels into the selected slot range.
8. Tap **Save to Radio** on the main screen to upload.

> *⚠ Imported channels fill **empty slots only** — existing channels are never overwritten. If there are more CSV entries than empty slots, a warning is shown and excess entries are skipped. The Starting Channel spinner is hidden in this case.*

Tone / TSQL Handling

The importer follows the CHIRP column spec:

Tone column	Behaviour
`Tone`	TX CTCSS only (`rToneFreq`); RX squelch is carrier-triggered
`TSQL`	TX + RX CTCSS using `cToneFreq` ; radio opens only when it hears the tone back
`DTCS`	TX + RX DCS using `DtcsCode` + `DtcsPolarity`
(blank)	No tone — **unless** `rToneFreq` ≠88.5 Hz (CHIRP's "no-tone" sentinel), in which case the frequency is imported as TX Tone encode-only

> 💡 ***RepeaterBook / database exports*** *commonly leave the Tone column blank while still populating* `rToneFreq` *with the repeater's access tone. The blank-column fallback above ensures that tone is captured as a TX encode rather than silently discarded. If you do not want the tone imported, zero out the* `rToneFreq` *column (or set it to* `88.5` *) before importing.*

Search RepeaterBook

Use ⋮ → **Search RepeaterBook…** after loading EEPROM from the radio. The **Search RepeaterBook** screen queries RepeaterBook, lets you multi-select repeaters, then **Import selected** opens the same **Import CHIRP CSV** flow as file/clipboard (groups, power, starting channel, preview).

What you need

- Your own RepeaterBook token, saved in the app (see Your RepeaterBook token below). The app does not contain a shared token.
- For **proximity (HTML) search** in the United States: set **Country** to **United States**, enter **non-zero latitude and longitude**, and a **distance** greater than zero (see Location and distance).

HOW THE SEARCH SCREEN WORKS

Field / control	Purpose
Country, **State / province**	Drives the JSON query when the app uses the **export API**. For **United States**, **Canada**, and **Mexico**, you must pick a **specific** state or province (not **All**) on that path — or use **US proximity** (lat/lon + distance) instead, which does not require state.
Service	**Amateur** or **GMRS** (GMRS is only enabled when country is United States).

Latitude / Longitude	Optional. With **United States** + both set + **distance** > 0, the app uses **HTML proximity** instead of export-only search (see below).
Use my location	Requests location permission, fills lat/lon when a fix is available. If **distance** is empty or zero, it is prefilled to **30** miles as a starting value (you can change it).
Distance limit miles	Radius for **US proximity** searches. RepeaterBook's HTML forms use `Dunit=m` (miles) for that radius; the value you enter is sent as miles. For **export API** results, the same field is converted internally for client-side distance filtering when coordinates are present.
Filter	Optional text filter (CHIRP-style: city, county, callsign, notes, etc.).
Open repeaters only	Keeps repeaters marked OPEN when data includes that field.
Convert dual-mode digital to FM	When importing, favors FM for dual-mode entries where applicable.

Results can be narrowed with the **Filter results** field after search. Attribution and API credit appear at the bottom of the screen.

EXPORT API VS. PROXIMITY (HTML) — WHEN EACH IS USED

The app uses **two different data paths**. Only one applies per search.

	Export API (`export.php` / `exportROW.php`)	Proximity (HTML)
Triggered when	Country is not *US+coords+distance*, or lat/lon/distance are missing, or distance is 0	**United States**, **non-zero lat/lon**, **distance > 0**
Amateur	JSON from `export.php` (NA) or `exportROW.php` (ROW) per the API wiki	HTML from **Proximity 2.0** `prox2_result.php` (default **2 m**, **FM** on the site form)
GMRS	Same JSON endpoints	HTML from `gmrs/prox_result.php`
Radius	If you provide lat/lon + distance, the app can **filter/sort** JSON rows by distance when records include coordinates (Haversine, miles converted to km internally). Otherwise state/country drive the query.	Radius is whatever RepeaterBook returns for your **miles** + `Dunit=m` request (same convention as the website).

Tones	Whatever fields exist in the JSON record (`PL` , `TSQ` , etc.)	List pages show a single **Access** column; the app can **follow each repeater's detail page** to read **Uplink Tone** / **Downlink Tone** and map them to **PL** / **TSQ** when available.
Token / HTTP client	Sends your personal token only in the `X-RB-App-Token` header on **export** requests. The token is never put in the URL.	Proximity pages are normal **HTML GET** requests. They do not receive the API token. The app still sends its User-Agent.

In short: **export API** = documented JSON export, best for **country/state** (and worldwide) workflows; **proximity** = same idea as using RepeaterBook's **location + radius** search on the web, for **US** users who enter coordinates and a **miles** radius.

REPEATERBOOK LOCATION AND DISTANCE

- **Proximity mode** requires **United States**, valid **latitude**, **longitude**, and **distance** (miles) **> 0**.
- **Use my location** helps fill coordinates; if distance was blank or zero, **30** miles is suggested so proximity mode can run without an extra step.

YOUR REPEATERBOOK TOKEN

This app is a distributed client. It does not ship a RepeaterBook credential, and it does not accept a shared application token.

1. Log in at RepeaterBook.
2. Open API Apps and generate a personal token for this app. It starts with `rbuapp_`.
3. In the app, open **Search RepeaterBook** and choose ⋮ → **RepeaterBook token**. Paste the token. It is stored encrypted on the device and is not shown again.
4. Search once. If RepeaterBook rejects the token, the scope, or the User-Agent, correct the token and try again once. Do not experiment with other authentication styles.

The app sends that token only as the `X-RB-App-Token` header on `https://www.repeaterbook.com/api/` requests. It never puts the token in a query string. On HTTP 429 it waits the full `Retry-After` time before another request.
If a token is exposed, revoke it on the API Apps page and create a new one. Do not send the token value by email.

PUBLIC APP IDENTITY (NOT A CREDENTIAL)

Optional entries in `local.properties` (next to the Android module) identify the app. They are not secrets, and they must not contain a token:

- `REPEATERBOOK_CONTACT_EMAIL=` — public contact email included in the User-Agent
- `REPEATERBOOK_APP_URL=` — public project or support URL included in the User-Agent
- `REPEATERBOOK_USER_AGENT=` — set this only when RepeaterBook has approved one exact User-Agent string. Do not change it while troubleshooting a rejected request.

Do not set a build-time token or an authentication-mode property. The build fails if either is present, or if one of the identity values contains a token. Rebuild after changing the public User-Agent identity.
North America queries use `export.php` (parameters include `state_id`, `county`, `emcomm`, `stype`); **rest of world** uses `exportROW.php` (parameters include `region` only — no `state_id`, `county`, `emcomm`, or `stype` in the documented ROW API). You can enter **multiple countries** separated by commas; the app sends repeated `country=` query arguments like the wiki's US+Canada example.
Data use is subject to RepeaterBook API terms. The search screen credits RepeaterBook and links back to RepeaterBook.com.

8.2 Sort Channels by Group

Automatically reorganize all channels so that each group's channels are contiguous in the channel list (useful for radios that scan group-by-group).

```
┌──────────────────────────────────────────────────┐
│ ← Sort Channels by Group                         │
├──────────────────────────────────────────────────┤
│  Drag groups into your preferred order.          │
│  Channels will be reorganized so each group's    │
│  channels appear together.                       │
│                                                  │
│  48 grouped across 5 group(s) · 12 ungrouped     │
│  · 138 empty slots                               │
│                                                  │
│  ≡  A  GMRS           (24 ch)                    │
│  ≡  B  MURS            (5 ch)                    │
│  ≡  C  Weather         (7 ch)                    │
│  ≡  D  Local          (12 ch)                    │
│  ≡  E  Simplex         (3 ch)  ← drag handle (≡) │
│  ≡  F  (no label)      (0 ch)                    │
│  …                                               │
│                                                  │
│  Ungrouped channels will be placed after         │
│  all group blocks. Empty slots always moved last.│
│                                                  │
│  [ Cancel ]              [ Sort ]                │
└──────────────────────────────────────────────────┘
```

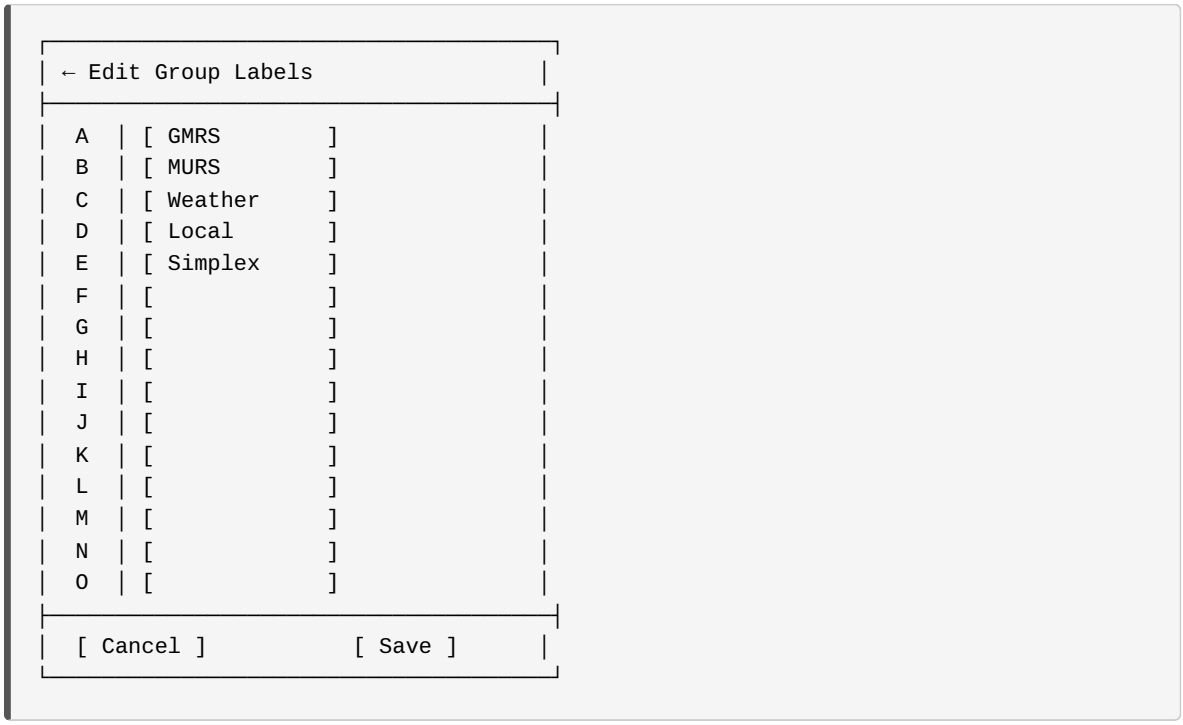
How it works

1. Drag group rows by their ≡ handle to set priority order.
2. Groups higher in the list get lower channel numbers after sorting.
3. Tap **Sort** — a confirmation dialog shows the final order before committing.
4. The sort is **stable**: channels within the same group keep their relative order.
5. Channels with multiple group assignments are sorted under their **highest-priority group** (the one earliest in your drag order).
6. Ungrouped channels are appended after all group blocks; empty slots go last.
7. Tap **Save to Radio** afterward.

> ***Example sort order:*** *A (GMRS) → B (MURS) → C (Weather) → Ungrouped → Empty Result: channels 1–24 = GMRS, 25–29 = MURS, 30–36 = Weather, 37–48 = ungrouped.*

8.3 Edit Group Labels

Assign human-readable names to the 15 group letters (A–O). Labels are stored in the EEPROM at offset `0x1C90` (6 bytes each, max 5 characters + null).



- Max **5 characters** per label (enforced by the text field).
- Labels appear throughout the app (channel editor group spinners, bulk group dialog, CHIRP import group spinners, sort screen).
- Tap **Save** — labels are written into the in-memory EEPROM and take effect in the app immediately. Upload to radio with **Save to Radio**.

8.4 Save / Import EEPROM Dump

Save EEPROM dump

Saves the currently loaded EEPROM as a raw `.bin` file to your device storage.

- The file is named `nicfw25_YYYYMMDD_HHmmss.bin`.
- Use this as a backup before making changes, or to share an EEPROM snapshot.

Import EEPROM dump

Loads a previously saved `.bin` file as the active EEPROM (bypasses Bluetooth).

- The file must be exactly 8192 bytes (8 KB).
- After import, the channel list refreshes as if you had loaded from the radio.
- Use **Save to Radio** to push the loaded dump to a connected radio.

> ***Workflow example:*** *Clone one radio's configuration to another: 1. Connect Radio A →* ***Load from Radio*** *→* ***Save EEPROM dump****. 2. Connect Radio B →* ***Import EEPROM dump*** *→* ***Save to Radio****.*

8.5 Edit Band Plan

Define up to 20 frequency band entries that control which frequencies are available for TX on the radio. nicFW 2.5 stores the Band Plan at EEPROM offset `0x1A00`.

```
┌──────────────────────────────────────────────────────────┐
│ ← Edit Band Plan                                         │
├──────────────────────────────────────────────────────────┤
│  Band plan controls which frequencies allow TX.          │
│  The radio's Band Plan overrides per-channel power.      │
├──────────────────────────────────────────────────────────┤
│  #1  │  136.000 – 174.000 MHz  │ TX ✓  FM  Wide  100     │
│  #2  │  400.000 – 480.000 MHz  │ TX ✓  FM  Wide  100     │
│  #3  │  108.000 – 136.000 MHz  │ TX ✗  AM  Narrow  0     │
│  #4  │  151.820 – 154.600 MHz  │ TX ✓  FM  Narrow  50    │  ← MURS
│  …                                                       │
│  #20 │  (empty)                                          │
├──────────────────────────────────────────────────────────┤
│  [ Cancel ]                    [ Save ]                  │
└──────────────────────────────────────────────────────────┘
```

Tap any row to edit that band plan entry:

```
┌──────────────────────────────────────────┐
│ ← Band Plan Entry #4                     │
├──────────────────────────────────────────┤
│  Start Freq (MHz)  [ 151.8200       ]    │
│  End Freq (MHz)    [ 154.6000       ]    │
│  Max Power         [ 50             ]    │
│  TX Allowed        [ ✓ Yes          ]    │
│  Modulation        [ FM  ▼          ]    │
│  Bandwidth         [ Narrow ▼       ]    │
│                                          │
│  [ Clear Entry ]                         │
│  [ Cancel ]          [ Save Entry ]      │
└──────────────────────────────────────────┘
```

Band Plan Field Reference:

Field	Description
Start / End Freq	Frequency range in MHz
Max Power	Raw power cap (0–255); 0 = Ignore (no cap applied)

TX Allowed	When off, the radio treats this range as RX-only
Modulation	Ignore / FM / AM / USB / Auto / Enforce variants
Bandwidth	Ignore / Wide / Narrow / FM Tuner

> *The radio applies Band Plan entries in order — the first matching entry wins. A catch-all entry (wide range, TX off) placed last covers all unlicensed frequencies.*

8.6 Edit Scan Presets

Configure up to 20 scan presets stored at EEPROM offset `0x1B00`. Each preset defines a frequency range for the radio's spectrum scanner.

```
┌──────────────────────────────────────────────────────────────┐
│ ← Edit Scan Presets                                          │
├──────────────────────────────────────────────────────────────┤
│  Scan presets define frequency ranges for the built-in       │
│  spectrum scanner. Up to 20 slots available.                 │
├──────────────────────────────────────────────────────────────┤
│  #1  | 108.00000 – 137.00000 MHz  "Air Band"                  │
│      |  Step: 25.00 kHz  Resume: 10  Persist: 0  AM  Ultra: 7 │
│  #2  | 137.00000 – 144.00000 MHz  "Satellite"                 │
│      |  Step: 25.00 kHz  Resume: 5   Persist: 0  FM  Ultra: 7 │
│  …                                                           │
│  #13 | 462.55000 – 462.75000 MHz  "GMRS"                     │
│      |  Step: 12.50 kHz  Resume: 0  Persist: 0  FM  Ultra: 7  │
│  …                                                           │
│  #20 | Empty — Tap to configure this slot                    │
├──────────────────────────────────────────────────────────────┤
│  [ Cancel ]                              [ Save ]            │
└──────────────────────────────────────────────────────────────┘
```

Tap any row to edit that scan preset slot:

```
┌──────────────────────────────────────────────┐
│ ← Scan Preset #1                             │
├──────────────────────────────────────────────┤
│  Start Freq (MHz)  [ 108.0000            ]   │
│  End Freq (MHz)    [ 137.0000            ]   │
│  Step (kHz)        [ 25.00              ]   │
│  Scan Resume       [ 10                 ]   │
│  Scan Persist      [ 0                  ]   │
│  Modulation        [ AM ▼]                   │
│  Ultrascan Speed   [ 7 ▼]                    │
│  Label (8 chars)   [ Air Band           ]   │
│                                              │
│  [ Clear Slot ]                              │
│  [ Cancel ]         [ Save Entry ]           │
└──────────────────────────────────────────────┘
```

Scan Preset Field Reference:

Field	Description
Start / End Freq	MHz — defines the scan range
Step	Scan step in kHz (e.g., 25.00 = standard FM, 12.50 = GMRS)
Scan Resume	Dwell count after signal lost before resuming scan
Scan Persist	Hold count when signal found
Modulation	FM / AM / USB / Auto (encoding matches EEPROM, not channel list)
Ultrascan Speed	0–7 (0 = slowest, 7 = fastest)
Label	Up to 8 ASCII characters, displayed on the radio screen

> ***Empty slot detection:*** *A slot is empty when* `startFreq == 0`*. Tapping* ***Clear Slot*** *zeros the entry. The EEPROM block has no magic header — each slot stands alone.*

8.7 Radio Settings

View and edit the nicFW Settings block stored at EEPROM offset `0x1900`. Access via ⋮ → **Radio Settings...** (requires EEPROM to be loaded).
Tap **Save Settings** to write all changes to the in-memory EEPROM, then tap **Save to Radio** from the main screen to upload to the device.

> *Fields marked ⚠ use EEPROM offsets or encodings that are not fully confirmed — treat them with extra caution.*

The editor is divided into ten sections:

Squelch & Audio

Field	Range	Default	Notes
Squelch	0–9 (spinner)	2	Squelch threshold level
Sq Noise Level	0–255	0	Raw noise floor threshold
Noise Ceiling	0–255	0	Upper noise limit for squelch detection
Sq Tail Elim	Off / On variants	Off	Squelch tail elimination mode
Mic Gain	0–31	25	Microphone preamp gain
Noise Gate	Off / On	Off	Enables the TX noise gate
RF Gain	AGC / 1–42	AGC	Receiver RF gain. AGC = automatic; 1–42 = fixed (lower = less gain)

Display

Field	Range	Default	Notes
LCD Brightness	0–35	28	Backlight level
LCD Timeout (s)	0–255 (0 = Off)	Off	Screen off delay in seconds
Heartbeat	0–30 (0 = Off)	Off	Idle animation speed
Dim Brightness	0–14 (0 = Off)	Off	Reduced brightness level after timeout
LCD Gamma	0–255	0	Display gamma correction
LCD Inverted	Off / On	Off	Inverts the display colors
S-Bar Style	Spinner	Segment	Signal bar display style
S-Bar Persistent	Off / On	Off	Keep signal bar visible when idle

TX Settings

Field	Range	Default	Notes
TX Mod Meter	Off / On	On	Shows TX modulation level on screen during transmit

TX Timeout (s)	0–255 (0 = Off)	120	Maximum continuous TX time before automatic cutoff
TX Sub Tone Deviation	0–127	64	Sub-audible tone deviation level for CTCSS/DCS
PTT Mode	Spinner	Dual	PTT button behavior (Dual = toggle + hold)
TX Filter Trans ⚠	0–65535	280.0	TX filter transition frequency (offset unconfirmed)
TX Current	Off / On	Off	Show transmit current on display
Repeater Tone (Hz)	0–65535	1750	1750 Hz burst tone for European repeater access

---

RX & Tuning

Field	Range	Default	Notes
Step	Spinner	5.0 kHz	VFO tuning step
AF Filters	Spinner	All	Audio filter combination
IF Freq	Spinner	8.46 kHz	Intermediate frequency
RFi Comp	Spinner	Off	RF interference compensation
AGC 0–3	0–63 each	24 / 32 / 37 / 40	AGC threshold levels (fine tuning)
AM AGC Fix	Off / On	Off	Corrects AM mode AGC behavior; recommended On when using AM channels

---

Scan

Field	Range	Default	Notes
Dual Watch	Off / On	On	Monitors two channels simultaneously
Scan Resume (s)	0–30	10	Seconds before scan resumes after losing a signal
Scan Range (MHz)	decimal	1.00	Frequency window scanned around current channel
Scan Persist (s)	decimal (0 = Off)	Off	Hold time when a signal is found

Scan Update (×0.1 s)	0–255	0	Display update interval during scan
Ultrascan	0–7 (0 = Off)	7	Spectrum scan speed (7 = fastest)

VOX

Field	Range	Default	Notes
VOX Level	0–15 (0 = Off)	Off	Voice-activated TX sensitivity
VOX Tail (s)	decimal	2.0	Seconds TX stays open after voice stops

---

Tones & Keys

Field	Range	Default	Notes
Tone Monitor	Spinner	On	Sub-audible tone monitor behavior
Key Tones	Spinner	Off	Button press beep style
Sub-Tone Deviation	0–255	74	Global sub-tone deviation level

DTMF

Field	Range	Default	Notes
DTMF Volume	0–255	80	Received DTMF audio level
DTMF Speed	0–15	11	DTMF tone transmit speed
DTMF Decode	Spinner	Off	DTMF decode / squelch mode
DTMF Seq End Pause (s)	decimal	1.0	Gap between DTMF sequence transmissions

---

System

Field	Range	Default	Notes
Battery Style	Spinner	Percent	Battery indicator format
Bluetooth	Off / On	On	Enable/disable the radio's Bluetooth module
Power Save	Off / 1–20	Off	Battery saving sleep interval. Higher = longer sleep cycles

DW Start Delay (s)	0–15	5	Dual Watch activation delay after squelch closes
DW VFO Lock	Off / On	Off	Lock VFO during Dual Watch mode
AllStar Link	Spinner	Off	AllStar VoIP repeater link mode
Disable FM Tuner	Off / On	Off	Hides the FM broadcast band from the tuner
Scrambler IF	Spinner	Off	Scrambler intermediate frequency; Off = disabled, 2600–3500 Hz in 100 Hz steps

---

Security

Field	Range	Default	Notes
PIN	0–9999	1234	Radio power-on or key-lock PIN
PIN Action	Spinner	Off	When PIN is triggered (power-on / key-lock / off)

8.8 Tune Settings

View and edit three per-radio calibration values stored at EEPROM offset `0x1DFB`. Access via ⋮ → **Tune Settings...** (requires EEPROM to be loaded).

> ⚠ ***These values are calibrated for your specific radio unit.*** *If you cloned another radio's EEPROM, recalibrate XTAL 671 and verify the power caps for the target unit before saving to that radio.*

Screen Layout



Field Reference

Field	EEPROM Offset	Range	Description
VHF Power Setting Cap	`0x1DFF`	0–255 (raw byte)	Maximum TX power for VHF frequencies (<300 MHz). Channels whose stored power exceeds this value are clamped by the radio at transmit time.
UHF Power Setting Cap	`0x1DFD`	0–255 (raw byte)	Same cap for UHF frequencies (≥300 MHz).

XTAL 671 Correction	`0x1` `DFB`	−128…+127 (signed)	Crystal oscillator correction applied by nicFW. Use the XTAL 671 Calculator to compute the correct value for your unit.

> *The live watt estimate displayed next to each cap picker (e.g., "≈ 5.0W") uses the same interpolation table as the channel list power column.*

How Power Caps Work

```
Per-channel power byte (stored in EEPROM)
        │
        ▼
  if value > cap  ──►  radio transmits at cap value
  if value ≤ cap  ──►  radio transmits at stored value
```

- **The stored EEPROM byte is never modified** — only the transmitted power is clamped.
- When a channel's stored power exceeds the cap, the app shows:
- A `⚠` suffix in the channel list power column (e.g., `5.0W ⚠`)
- A non-blocking advisory below the Power spinner in the channel editor
- A non-blocking advisory below the wheel in the bulk TX power picker
- **Raising the cap later** automatically un-clamps all affected channels without requiring any per-channel edits.

Channel List Indicators

Power display	Meaning
`5.0W`	Normal — within cap and TX allowed
`5.0W ⚠`	Stored power exceeds VHF/UHF cap; radio will clamp at TX time
`5.0W (BP)`	Band Plan marks this frequency as TX-restricted
`5.0W (BP) ⚠`	Both Band Plan TX-restricted AND power exceeds cap
`N/T`	No-transmit (power = 0 or Band Plan TX off)

Protect Tune Settings

A checkable toggle in the overflow menu (⋮ → **Protect Tune Settings**) that locks the per-radio calibration values and enables a safe multi-radio write workflow.

The same toggle is also available inside the Tune Settings screen itself as a switch at the top of the page.

> ***Default on new installs:*** *Protect Tune Settings starts* ***ON*** *by default. Existing installs keep their previously saved preference.*

When protection is OFF: - Tune Settings fields (XTAL, VHF cap, UHF cap) are fully editable. - **Save to Radio** uploads the EEPROM as-is — calibration bytes are whatever is in the current template.
When protection is ON: - The XTAL, VHF cap, and UHF cap pickers are greyed out and non-editable. - The **Save Tune Settings** button is disabled. - **Save to Radio** performs a two-phase protected write:

```
Phase 1 — Pre-read target radio
  Progress: "Reading target radio calibration…"
  Downloads the connected radio's full EEPROM
  Extracts the 5 calibration bytes (0x1DFB–0x1DFF)

Phase 2 — Patched upload
  Progress: "Writing…"
  Uploads a copy of the template EEPROM with those 5 bytes
  substituted from the target radio's own calibration
  Template in memory is never modified
```

Multi-Radio Template Workflow

The typical use case for Protect Tune Settings is writing the same channel/settings configuration to a collection of radios while preserving each radio's factory calibration:

1. Connect **Radio 1** → **Load from Radio**.
2. Edit channels, settings, and band plan as desired. Do **not** edit Tune Settings.
3. Enable **Protect Tune Settings** (✓ appears in the menu).
4. Disconnect Radio 1, connect **Radio 2**.
5. Tap **Save to Radio** → confirm dialog shows 🔒 protection note.
6. Progress shows "Reading target radio calibration..." then "Writing...".
7. Radio 2 receives the updated channels/settings but **keeps its own XTAL and power caps**.
8. Repeat steps 4–7 for Radio 3, 4, ... — each radio keeps its own calibration.

> ***Template file workflow:*** *Save an EEPROM dump (⋮ →* ***Save EEPROM dump...****) while protection is enabled. Load that* `.bin` *file later and apply it to any number of radios — each will retain its own calibration on write.*

> ***CHIRP note:*** *The Protect Tune Settings feature is* ***Android-only****. The CHIRP driver does not implement a pre-read, so it always writes whatever calibration bytes are in the loaded image. Use the Android app for multi-radio template workflows.*

Workflow

1. Tap ⋮ → **Tune Settings…**
2. Scroll the **VHF Power Setting Cap** picker to your desired ceiling raw value.
3. Example: `130` ≈ 5.0 W (matches the field default seen in the NICFW Programmer).
4. Set `255` to disable the cap (allow full power on all channels).
5. Repeat for **UHF Power Setting Cap**.
6. Adjust **XTAL 671** if needed — use the XTAL 671 Calculator to derive the correct value first.
7. Tap **Save Tune Settings** → values are written to the in-memory EEPROM.
8. Tap **Save to Radio** on the main screen to upload.

> *After saving Tune Settings and returning to the channel list, scroll the list to verify that* `⚠` *indicators appear or disappear as expected.*

8.9 XTAL 671 Calculator

A standalone frequency calibration tool — no EEPROM or radio connection required. It computes the **nicFW XTAL 671 correction value** needed to compensate for crystal oscillator error in the TD-H3.

> ***Available any time*** *from the overflow menu, even before loading from the radio.*

⚠ Prerequisite — Reset XTAL to 0 First

> ***Before measuring the actual RX frequency you must set the radio's XTAL 671 value to 0.*** *A non-zero XTAL setting shifts the radio's receive frequency, which will corrupt the measurement and produce a wrong correction value.*

Steps: 1. On the radio, open Settings → XTAL 671 and set it to **0**. 2. Tune the radio to the target frequency. 3. Measure the actual received frequency with a calibrated reference (SDR, spectrum analyzer, or frequency counter). 4. Enter that measured frequency in the calculator.

Calculator Screen



How to Use

1. Tap ⋮ → **XTAL 671 Calculator...** from the main screen (available at any time).

2. Tap a **preset chip** to fill the target frequency, or type it manually.
3. Enter the **Actual RX Frequency** you measured with your reference device.
4. The **correction value** updates live as you type.
5. Tap **Copy Value to Clipboard** then enter it in the radio's XTAL 671 setting.

Formula

```
XTAL671 = ROUND( ((Actual − Target) × 100000) ÷ Target × 671.08864, 0 )
```

This matches the Google Sheets reference formula: `=ROUND(((D2-C2)*100000)/C2 * 671.08864, 0)` where C2 = Target Freq (MHz), D2 = Actual RX Freq (MHz).

Interpreting the Result

Result	Meaning	Action
Positive (e.g. +29)	Radio receives HIGH — oscillator running fast	Set XTAL 671 to +29
Negative (e.g. −35)	Radio receives LOW — oscillator running slow	Set XTAL 671 to −35
Zero	Within calibration tolerance	No change needed

Verification Example (from reference spreadsheet)

Radio	Target (MHz)	Actual RX (MHz)	XTAL 671
xxx579	462.725	462.7252	**29**
xxx591	462.725	462.72524	**35**
xxx866	462.725	462.7252	**29**
xxx869	462.725	462.72528	**41**
xxx589	462.725	462.72532	**46**

> *Each radio has a unique crystal tolerance; calculate and set individually for each unit.*

Preset Frequencies

Chip	Frequency	Band
GMRS22 462.725	462.7250 MHz	GMRS channel 22 (most common reference)
GMRS1 462.5625	462.5625 MHz	GMRS channel 1
MURS1 151.820	151.8200 MHz	MURS channel 1

WX1 162.400	162.4000 MHz	NOAA Weather channel 1

---

9. CHIRP Adapter (Python Driver)

The repository also includes a CHIRP radio driver: `tidradio_h3_nicfw25.py`
This driver lets you use **CHIRP** (the desktop radio programming software) to program the TD-H3 from a **PC** over a USB serial connection (connect the radio's USB-C port directly to your PC — no separate programming cable required). This is a desktop/PC connection and is separate from the Android app, which uses Bluetooth BLE.

Connection

1. Open CHIRP.
2. Select `Radio → Download from Radio`.
3. Choose **TIDRadio TD-H3 (nicFW 2.5)** from the radio list.
4. Select the correct COM port / serial device.
5. Click OK — CHIRP reads the full EEPROM via the same 0x45/0x30/0x46 protocol.

CHIRP Tab Overview

The driver exposes three extra setting tabs alongside the standard channel grid:

Band Plan Tab

Mirrors the Android **Edit Band Plan** screen. Each of the 20 band plan entries is shown as a CHIRP settings group with fields for start/end frequency, max power, TX allowed, modulation, and bandwidth.

Scan Presets Tab

Mirrors the Android **Edit Scan Presets** screen. All 20 scan preset slots are editable with the same fields: start/end frequency, step, resume, persist, modulation, ultrascan speed, and label.

Group Labels Tab

Mirrors the Android **Edit Group Labels** screen. Displays 15 text fields (A–O) for assigning human-readable names to channel groups.

Supported Channel Fields

CHIRP Field	EEPROM Field
Frequency	freqRxHz
Offset	offsetHz (duplex mode)
Duplex	duplex (+/−/split/none)
Name	name (12 chars)
Power	power (raw 0–255, mapped to Low/Medium/High/Turbo)
Mode	mode (FM/AM/USB)
Tone Mode	txToneMode (Tone/DTCS/none)
CTCSS / DCS	txToneVal + txTonePolarity

---

10. Tips & Troubleshooting

General Workflow

```
Load from Radio
      │
      ▼
  Edit channels / Band Plan / Scan Presets / Groups
      │
      ▼
  [Save EEPROM dump]  ← optional backup before upload
      │
      ▼
Save to Radio
```

Always back up your EEPROM dump before making major changes.

Common Issues

Issue	Resolution
Connect button grayed out	Not connected — tap Connect first
Load / Save buttons grayed out	No Bluetooth connection established

"Frequency out of range" on save	Frequency is outside Band Plan entries; verify band plan or use a supported frequency
"No empty slots" on CHIRP import	All 198 channels are used; clear some channels first
Channel list shows nothing after load	EEPROM read may have failed; try disconnecting and reconnecting, then load again
Scan Preset entry shows 1.0 MHz	Slot is empty (`startFreq == 0`); this is a display artifact in some tools — empty slots are handled correctly by nicFW
Channel list shows ⚠ next to power	Stored channel power exceeds the VHF or UHF cap in Tune Settings. Lower the channel power OR raise the cap in ⋮ → **Tune Settings…**
BLE scan finds no devices	Power on the radio; disconnect other phones; grant **Nearby Devices** (Android 12+) or **Location** (older). **v1.1.0+:** only devices that **advertise** a supported UART service UUID appear — if yours does not, use **Classic BT** or check nicFW BT advertising
BLE connects then misbehaves on a dongle	**v1.1.0+** uses safer MTU/chunk sizing; update to the latest app build. If problems persist, try Classic SPP or another adapter
Classic BT connection fails	Pair the radio in Android Bluetooth settings first, then retry

---

Power Estimate Reference

Use this quick table when setting power in bulk:

```
Target   Raw Value (approx.)
──────────────────────────────
 N/T          0  (N/T)
 0.5 W       29
 1.0 W       44
 2.0 W       58
 3.0 W       86
 5.0 W      130
 Max        255
```

Group Design Tips

- Use **Group A–D** for operational categories (e.g., GMRS, MURS, Weather, Simplex).
- Assign channels to **multiple groups** when they logically belong to more than one category (e.g., a GMRS repeater might be in Group A *and* Group D for local repeaters).

- After organizing groups, run **Sort Channels by Group** to pack related channels into contiguous slot numbers for cleaner scanning.
- Use **Bulk Edit** (✎ in multi-select mode), then choose **Channel groups**, to assign a group to dozens of channels at once instead of editing each one individually.

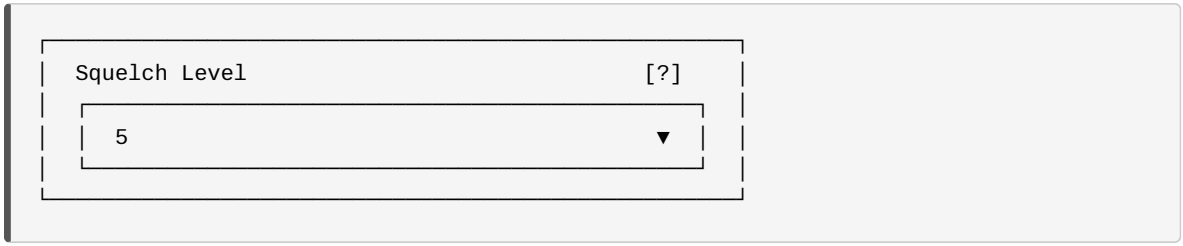
Keyboard / Navigation

Action	How
Enter multi-select	Long-press a channel row
Exit multi-select	Tap Done in the action bar
Scroll channel list	Swipe up/down on the list
Reorder a channel	Drag the ≡ handle (outside multi-select mode)
Go back from any editor	← toolbar back arrow or Android back gesture
Open context help for a field	Tap the **?** (lightbulb) icon next to the field label

---

11. Context Help

Every setting in the **Radio Settings**, **Channel Editor**, and **Tune Settings** screens has a built-in help button: a small **?** (lightbulb) icon next to the field label.



Tapping **?** opens an AlertDialog with:

```
┌──────────────────────────────────────────────────┐
│  Squelch Level                                   │
├──────────────────────────────────────────────────┤
│  Range: 0–9  |  Default: 5                       │
│                                                  │
│  Controls the squelch threshold. Lower values open │
│  squelch at weaker signals. 0 = squelch always   │
│  open (carrier squelch disabled).                │
│                                                  │
│  Note: Applies to the currently selected channel │
│  memory bank.                                    │
│                          [ OK ]                  │
└──────────────────────────────────────────────────┘
```

Dialog fields:

Field	Content
Title	Setting name
Range	Valid value range or options
Default	Factory default value
Description	Plain-language explanation of what the setting does
Note	Additional context, caveats, or interactions with other settings (when applicable)

Where context help is available:

Screen	Help available on
Channel Editor	All fields: Frequency, Duplex, Name, Power, Mode, Bandwidth, Busy Lock, TX Tone, RX Tone, Groups
Radio Settings	All 56 settings across all 10 sections
Tune Settings	VHF Power Cap, UHF Power Cap, XTAL 671 Correction

Help content is sourced from the nicFW V2.5 manual and stored in `docs/help_reference/help_content.yaml` in the repository.

End of User Guide